METRIC

MIL-PRF-6086G
26 March 2025

SUPERSEDING
MIL-PRF-6086F
W/AMENDMENT 1
15 May 2014

PERFORMANCE SPECIFICATION

LUBRICATING OIL, GEAR, PETROLEUM BASE

(NATO O-153, O-155)

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades of petroleum-based lubricating oils for gears (see 6.1). This oil is identified by NATO Code Numbers O-153 and O-155 (see 1.2).

1.2 Classification. The oil covered by this specification is classified as one of the following two grades:

a. Grade L – Light; identified by military symbol OGL and NATO Code O-153.

b. Grade M – Medium; identified by military symbol OGR and NATO Code O-155.

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE FORMS MANAGEMENT PROGRAM

DD Form 1718 - Certification of Qualified Products

(Copies of this document are available online at https://www.esd.whs.mil/DD/.)

DEFENSE STANDARDIZATION PROGRAM OFFICE

SD-6 - Provisions Governing Qualification

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D91 - Standard Test Method for Precipitation Number of Lubricating Oils

ASTM D92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester

ASTM D97 - Standard Test Method for Pour Point of Petroleum Products

ASTM D130 - Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test

ASTM D445 - Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)

ASTM D892 - Standard Test Method for Foaming Characteristics of Lubricating Oils

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ASTM D974	-	Standard Test Method for Acid and Base Number by Color-Indicator Titration
ASTM D1500	-	Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale)
ASTM D2270	-	Standard Practice for Calculating Viscosity Index from Kinematic Viscosity at 40 and 100°C
ASTM D2783	-	Standard Test Method for Measurement of Extreme-Pressure Properties of Lubricating Fluids (Four-Ball Method)
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D5949	-	Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method)
ASTM D5950	-	Standard Test Method for Pour Point of Petroleum Products (Automatic Tilt Method)

(Copies of these documents are available from https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The gear oil furnished under this specification shall be products that are authorized by the qualifying activity for listing on the Qualified Products List QPL No. 6086 before contract award (see 4.2 and 6.4). Changes shall not be permitted in the formulation of an approved product unless specific, written approval of the qualifying activity is obtained.

3.2 Materials. The oil shall consist of base stock oil mixed with a suitable load-carrying additive. The additive shall not be corrosive or cause excessive foaming, and shall be completely compatible with the base stock. The oil shall be clear and free of visible water, suspended matter, dirt, sediment or other impurities when observed at room temperature (25 °C ±2 °C). Recycled constituent materials shall not be used.

3.3 Chemical and physical and performance requirements. The finished gear oil shall conform to the requirements specified in Table I.

3.4 Toxicity. The gear oil shall have no adverse effect on the health of personnel when used for its intended purpose (see 6.6).

3.5 Workmanship. The oil shall be entirely homogeneous and free of lumps undissolved additive, water, dirt, lint, or sediment.

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TABLE I. Properties of finished fluids.

Property	Requirements, Grade L	Requirements, Grade M	Test Method
Viscosity, mm^2/s @37.8 °C	23-34	60-82	ASTM D445
Viscosity index, (min)	80	80	ASTM D2270
Flash point, °C (min)	137.8	154.5	ASTM D92
Pour point, °C (max)	- 40	- 28.9	ASTM D97[1], D5949, or D5950
Load wear index, kg (min)	40	40	ASTM D2783
Precipitation number, mL (max)	0.1	0.1	ASTM D91
Acid number, mg KOH/g (max)	1.0	1.0	ASTM D974
Color, (max)	8	8	ASTM D1500
Foaming Characteristics (tendency, stability)			ASTM D892
Sequence I, mL (max)	10, 0	10, 0	
Sequence II, mL (max)	20, 0	20, 0	
Sequence III, mL (max)	10, 0	10, 0	
Copper corrosion (3 hours @ 100 °C), (max)	2	2	ASTM D130

NOTES:

1. Referee Method.

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4. VERIFICATION

4.1 Classification of inspections. The examination and testing of the finished gear oil are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall consist of testing to all the requirements specified in sections 3.3 through 3.5 and Table I. When required by the qualifying activity, additional evaluations may be required on candidate formulations.

4.2.1 Qualification process. At the initiation of the qualification process, prospective suppliers will forward a written request for such action to the qualifying activity, Air Force Petroleum Office, AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or emailed to AFPET.PTPS@us.af.mil. This request will contain general information on the proposed candidate material. The qualifying activity will reply in writing outlining the required information to be submitted, the governing regulations for qualified products, and the cost of testing. The information that will be submitted includes:

a. Letter stating the company will adhere to the appropriate provisions of the SD-6 on company letterhead.

b. Complete formulation of the candidate product, including chemical composition in weight percent, the manufacturer, the commercial trade name and CAS number, the purities, and use of each component.

c. Safety Data Sheet (SDS) for each component and final product. The SDS will not be older than 5 years.

d. Certified laboratory test data from the manufacturer, a commercial laboratory, or combination thereof showing quantitative results of all qualification tests required by this specification as identified in Table I.

e. Verification that the System for Award Management (SAM) (CAGE Codes for the corporate office and each plant) is current. This can be viewed at https://sam.gov/.

f. Identification of the manufacturing site of the specific batch of test sample to be submitted.

g. A completed DD Form 1718.

4.2.2 Qualification inspection sample. A qualification sample shall consist of a one (1) gallon container of gear oil. The qualifying activity will request manufacturer and third-party data prior to authorizing submission of the qualification sample.

4.2.3 Retention of qualification. To retain qualification of products approved for listing on the Qualified Products List QPL No. 6086, the manufacturer will be requested to verify by certification (DD Form 1718) to the qualifying activity that its product(s) comply with the requirements of this specification. The manufacturer shall also provide a recent Certificate of Analysis (COA), to include all conformance inspection requirements (see 4.3), a current SDS, and formulation. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification will be in 2 1/2 year intervals from the date of original qualification

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and will be initiated by the manufacturer. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.2.4 Requalification. Requalification shall be required every five (5) years or when any change is made in the source of manufacture, purity, or composition of the fluid's base stock or additives. A minor change in the fluid formulation such as a new supplier of stock material might not need requalification testing, but only after notification to, and approval by the qualifying activity (see 6.3). The documentation required for requalification are the same as requirements stated in section 4.2.1 and shall include product formulation, SDS for each component and final product, a COA showing compliance to all of the qualification requirements specified in Table I and a one (1) gallon requalification sample from a recent production run within the previous 60 days. The SDS will not be older than five (5) years. The qualifying activity may, at its discretion, waive complete requalification or may require only partial requalification testing to determine the significance and acceptability of the proposed formulation change.

4.3 Conformance inspection. The conformance inspection shall be conducted on samples from randomly selected containers of each packaged lot (see 6.4.2) and consists of tests to verify conformance to the specific conformance requirements in Table I and sections 3.3 thru 3.5. Samples shall be labeled completely with information identifying the purposes of the sample, name of product, specification number, lot and batch number, date of manufacture, date of sampling, and contract number. Sampling shall be performed in accordance with ASTM D4057.

4.4 Method of inspection. Tests shall be performed in accordance with the methods specified in Table I and section 4. Physical and chemical values, specified in section 3, apply to the arithmetic average of the determinations made on the samples for those values which fall within any stated repeatability or reproducibility limits of the applicable test method.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the Military Service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The two grades of gear oil covered by this specification are intended for use in light-to-moderate-loaded gear trains or gear boxes that operate in low-to-medium-speed ranges down to -40 °C.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grade and quantity of oil required.

c. Packaging requirements (see 5.1) to include size and type of container, level of packaging and level of packing required.

6.3 Qualification. With respect to products requiring qualification, contract awards will be made only for such products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 6086 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433; e-mail: AFPET.PTPS@us.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification information. It is understood that the material furnished under this specification subsequent to final approval will be of the same composition and will be equal to products on which approval was originally granted. In the event that fluid furnished under contract is found to deviate from the composition of the approval product, or that the product fails to perform satisfactorily, approval of such products may be subjected to immediate withdrawal from the QPL at the discretion of the approving activity.

6.3.2 Proceed authorization notification. Upon successful review of the submitted information, the Air Force Petroleum Office will issue authorization to ship the qualification sample(s) for testing.

6.3.3 Qualification samples. Upon receiving authorization from the Air Force Petroleum Office, qualification sample(s) may be shipped to the qualification activity, as detailed in the authorization letter. Sample(s) should be accompanied by an SDS and a copy of the authorization letter. Each sample should be plainly identified by securely attached durable tags or labels marked with the following information:

a. QUALIFICATION SAMPLE

b. Lubricating Oil, Gear, Petroleum Base

c. MIL-PRF-6086

d. Grade M or L

e. Name of Manufacturer

f. Product Code Number

g. Batch or Lot Number

h. Date of Manufacture

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6.4 Definitions.

6.4.1 Packaged lot. A packaged lot is an indefinite number of unit containers of identical size and type offered for acceptance and filled with a homogeneous material mixture from one isolated container; or filled with a material mixture manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.5 International standardization agreement implementation. This specification implements:

a. NATO STANAG/AFLP-1135 *Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations*.

When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the International Standardization Agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.6 Toxicity. Questions pertinent to this effect should be referred by the contracting activity to the appropriate departmental medical service who will act as its advisor.

6.6.1 Safety Data Sheets. Contracting officers will identify those activities that require copies of completed SDS prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume 1, *Shelf-Life Management Program: Program Administration*; and Volume 2, *Shelf-Life Management Program: Materiel Quality Control Storage Standards*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.8 Subject term (key word) listing.

Flash point
Foaming characteristics
Gear boxes
Gear trains
Viscosity

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6.9 Conversion of metric units. Units of measure have been converted to the International System of Units (Metric) IAW IEEE/ASTM SI10 *American National Standard for Metric Practice*. If test results are obtained in units other than Metric or there is a requirement to report dual units, IEEE/ASTM SI10 or NIST SP 811 *Guide for the Use of the International System of Units (SI)* should be used to convert the units.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – AS
Air Force – 68
DLA – GS

Review activities:
Army – AT, MI, MD1
Navy – SA, SH
Air Force – 20
DLA – PS

Preparing activity:
Air Force – 68
(Project 9150-2024-010)

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